

Prospectus

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Brief background to dark matter

In 1980, Vera Rubin published a paper that is credited with providing the first substantial evidence for dark matter (Rubin, Ford, and Thonnard 1980). However, no one has been able to identify the substance beyond speculation. Reasonable efforts have failed to describe the rotation curves Rubin first noticed, which implies that a new direction needs to be tried. Modern plasma physics gives insight on what to try.

Background to research

When a photon travels through a plasma medium, its energy is conserved. This means that the plasma frequency and wave number of the photon must change inversely, where the plasma frequency is dependent on the density of the plasma. This implies that as a photon travels through a plasma with varying density, if the plasma density goes up, the wave number of the photon must go down. Using Snell's law, the photon's angle must increase away from the path of travel, meaning that through a plasma gradient it diverges from its initial path. This behavior is based on the equation:

$$\omega^2 = \omega_p^2 + k^2 \quad (\text{natural units assumed}),$$

which is similar in form to Einstein's equation:

$$E^2 = m^2 + p^2.$$

This implies that we may be able to treat mass as a variable and obtain a similar effect. However, the treatment is more complex, as the momentum is inherently mass dependent and the opposite dependence is necessary for the observed galactic curves.

My planned research

A detailed treatment of the one-dimensional case can be found in (Shi 2021). However, the interaction of orbital bodies is two-dimensional, and work needs

to be done to extend the research to two dimensions. This can be done by modifying the Schwarzschild metric to encompass a non-constant mass distribution.

The one-dimensional case gave a differential equation for the mass distribution, and using a known trajectory, I was able to find a mass distribution that agreed with that trajectory. The two-dimensional case is inherently more difficult. Although an analytic solution to the Schwarzschild geodesic equation exists (Hackmann and Lämmerzahl 2008), it is much more common to use numerical solutions because of the complexity of the system.

In addition to having the potential to describe the observed behavior of sparse matter in galaxies, this approach offers great flexibility and if there is an analytic form of the mass distribution is guaranteed by the structure of the method to yield an analytic solution. It is likely to be as complex as, or more complex than, a Schwarzschild geodesic equation, but will reduce the method to an ODE. If an analytic mass distribution exists, this will allow for a general comparison to observed data and theoretical results, meaning that if uniqueness can be proven for the result, it will either be a solution to all observed trajectories or to only one observed trajectory (or a small subset contingent on observational error).

Timeline:

November-December: Finished interview problem

January-February: Solved a modified version of interview problem for mass distribution based on trajectories found in interview problem. Proved uniqueness.

March-April: Began working on preliminary math to start working on the orbital case. I found a non-relativistic form of the modified potential.

May: I aim to have a form of the mass distribution (not necessarily analytic) and a trajectory to match it to.

June-July: I aim to have an idea of whether an analytic form of the mass distribution is possible, and have a body of papers containing observational data to compare the model to. If the solution I find is analytic, I plan to have uniqueness proved or disproved at this point.

August: I hope to have the bulk of my thesis writing done by the end of August.

References

Cited material

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Supplementary material

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